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REVIEW

Safety of beta-alanine supplementation in humans: a narrative review

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Abstract
Carnosine levels play a significant role in intracellular pH buffering during exercise. The limiting factor for muscle carnosine synthesis is the availability of β-alanine. Results from meta-analysis showed a significant positive effect of β-alanine supplementation on high-intensity exercise performance. Nonetheless, much less has been described about the safety of β-alanine. The available literature indicates no adverse events related to β-alanine usage; side effects such as paresthesia may be observed if a large single high dose of β-alanine is taken, but the symptom can be attenuated either by using splitting doses (< 1.6 g) or a sustained-release formula. No adverse effects have been reported up to 24 weeks of β-alanine supplementation (3.2 $g.day^{-1}$), but the adverse effects of longer supplementation periods are still unknown. Most of the existing clinical studies were not designed with safety evaluations as primary end points. β-Alanine is currently considered to be safe in healthy populations, at recommended doses, by the International Society of Sports Nutrition stands position, while the Australian Institute of Sports has established β-alanine as a safe performance-enhancing supplement with strong scientific evidence (grade A). Cohort studies are needed evaluating the safety of β-alanine among representative populations, and the effects of co-variables such as sex, age, and ethnicity.

Keywords Safety · Adverse events · Side effects · Exercise · Performance

Introduction

Glycolysis is the primary energy source to sustain muscle contraction during high-intensity exercise. Knowingly, glycolysis is paralleled by hydrogen cation (H^+) production; when glycolytic rates are high, the rate of H^+ appearance exceeds the capacity of muscle cells to neutralize them, leading to the accumulation of H^+ in sarcoplasm as exercise progresses. Intramuscular pH consequently falls from ~ 7.1 at rest to ~ 6.5 at the point of exhaustion [1]. Muscle acidosis has been shown to interfere with several metabolic and contractile processes that result in reduced force production, slowed contraction and, ultimately, fatigue [2]. Although acidosis is not the sole cause of peripheral fatigue, intramuscular H^+ accumulation has been shown to disrupt phosphoryl creatine resynthesis [3], inhibit key enzymes of glycolysis [4] and impair contractile processes [5, 6].

Carnosine (β-alanyl-L-histidine) is a dipeptide synthesized in skeletal muscle from the amino acids L -histidine and β-alanine that has been shown to play an important role in intracellular pH regulation [7]. The rate-limiting factor for carnosine synthesis is the low availability of β-alanine [8–10]. Thus, β-alanine supplementation increases carnosine content in muscle cells [10] and its supplementation can improve performance in high-intensity exercise [11, 12]. However, although the effects of β-alanine supplementation on performance has been extensively studied [11, 12], the safety of this supplementation is less known [13]. Therefore, the aim of this narrative review is to discuss the safety of β-alanine supplementation on health parameters. We also aim to briefly review the current literature regarding the efficacy of β-alanine supplementation on performance as well as its biological functions.

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Physiological roles of carnosine

There is a growing interest to unravel the properties and potential applications of carnosine and related dipeptides (i.e., methylated analogs, anserine and ophidine, and acetylated analogs, n-acetylcarnosine and *N*-Acetylserine). Most of the fundamental research on carnosine was in the field of biochemistry/molecular biology, pharmacology and physiology, with applications ranging from neurology to diabetes, cardiovascular disease, nutrition, sports sciences and other areas [14].

Intracellular pH regulation

Carnosine belongs to a group of physicochemical buffers that provide the first line of defense against local changes in intracellular pH. The H^+ buffering properties of carnosine arise from the pKa of its imidazole ring (6.83) [14]. Carnosine plays a relevant pH-regulation role in the exercising skeletal muscle, where carnosine is most abundant—typical values range from ~ 15 mmol mmol·kg^{-1} dry muscle in vegetarians [15] to more than 40 mmol·kg^{-1} dry muscle after β-alanine supplementation [16]. Recent evidence indicates that carnosine can also play important pH-regulation roles in the cardiac muscle, especially when intracellular pH drops during acute ischemia [17], thereby conferring protection against ischemic injury. Nitrogen atoms on the imidazole ring can readily accept a proton at physiological pH, and therefore it has been suggested that carnosine buffering precedes the involvement of the bicarbonate buffering system during exercise [18]. However, the contribution of carnosine to the total muscle buffering capacity is approximately 7% [19], although this can be ~ 20% in type II fibers following β-alanine supplementation [10].

Other roles of carnosine

The fact that carnosine is abundant in tissues other than skeletal muscle suggests that the potential physiological roles of carnosine extend beyond its function as a proton buffer. In fact, numerous other roles for carnosine have been purported, such as protection against oxidative damage, protein glycation, Ca^{2+} buffering, and regulation of Ca^{2+} transients and Ca^{2+} sensitivity [20].

It has been suggested that excess of reactive oxygen species (ROS) may contribute to muscle fatigue and exercise-induced muscle damage [21–23]. Carnosine can act as an antioxidant by scavenging free radicals and singlet oxygen [24, 25], thereby having potential to attenuate oxidative stress. Carnosine can further reduce oxidative stress by chelating transition metals, such as copper and iron [25]. In doing so, these transition metals are prevented from reacting with peroxides in the Fenton reaction, which may further reduce free radical production. Hence, in addition to the well-demonstrated ergogenic effects, carnosine is also thought to have therapeutic benefits [26], although its use in disease prevention and treatment merits further investigation.

Carnosine displays a broad antioxidant activity, likely reflecting the properties of its imidazole ring. This can serve efficiently as an electron donor, preventing lipid peroxidation. Recent studies have provided experimental evidence for the role of carnosine in the detoxification of toxic products of lipid peroxidation in humans, such as reactive aldehydes [27, 28]. Such a role has also been shown to occur in the skeletal muscle in response to exercise [27]. Carnosine also quenches singlet oxygen and interacts with superoxides, improving their molecular stability and reducing reactivity [25, 29].

Carnosine metabolism

The rate-limiting factor for carnosine synthesis is, under normal conditions, the low availability of β-alanine [8–10]. β-Alanine is a nonessential non-proteogenic amino acid, naturally present in beef, pork, poultry and fish that can be endogenously synthesized in the uracil degradation pathway [30, 31], although at very low rates. Circulating β-alanine can be transported into tissues, such as the skeletal muscle, through two transporting proteins, namely TauT and PAT1. Once inside skeletal muscle cells, β-alanine can be used for carnosine synthesis or degraded by transaminases and then directed toward oxidation [32].

Carnosinase, the enzyme that hydrolyzes carnosine, is highly active in the digestive tract and in the blood, meaning that most of the intact dipeptide is degraded into its constituents' amino acids before reaching the tissues. In the skeletal muscle, however, a far less active isoform of carnosine is expressed, which seems to explain the high stability of carnosine within this tissue [33].

β-Alanine supplementation in exercise

Although carnosine properties have attracted some interest from researchers over the past century, the focus has shifted to the exercise science in the last decades after the seminal study by Harris et al. [10], which showed that chronic intake of β-alanine can increase muscle carnosine in human muscle, with following studies showing improved exercise capacity [34].

Supplementation with β-alanine has been shown to significantly increase carnosine levels in both type I and type

II human muscle fibers of the vastus lateralis [10], as well as in the tibialis anterior, soleus, gastrocnemius and deltoid [35–38]. β-Alanine supplementation for 4 weeks or more at ~ 3.0–6.4 g.day^{-1} has been shown to elevate intramuscular carnosine concentration by ~ 30–100% in a dose-dependent fashion [10, 34–36, 39, 40]. Dosing regimens as low as 1.6 g/day for 2 weeks may also increase muscle carnosine [33]. Although the magnitude of carnosine accretion in skeletal muscle is linearly correlated with the total dose of β-alanine [33], this appears to occur only during the first few weeks of supplementation. As the total accumulated dose increases, muscle carnosine seems to approach saturating concentrations and thus displays a curvilinear with a tendency to plateau [16]. A mathematical model that predicts carnosine increases over the course of β-alanine supplementation describes a logarithmic growth where saturation may limit carnosine to further accumulate after a certain point [41]. Nonetheless, there exists a large inter-individual variability in the muscle carnosine responses to β-alanine supplementation and some individuals may be able to accumulate large amounts of carnosine without showing signs of saturation [12].

The effects of β-alanine supplementation on exercise capacity and performance

Several meta-analyses showed a significant positive effect of β-alanine supplementation on exercise capacity, supporting the notion that β-alanine is an effective ergogenic aid [11, 12, 42]. Improvements in exercise capacity have been shown at doses ranging from 3.2 to 6.4 g/day (typically split into 3–4 single doses of 0.8–1.6 g) [43] for 4–24 weeks [34, 44]. The magnitude of changes in exercise capacity has been associated with the magnitude of changes in muscle carnosine [34, 45]. Importantly, total accumulated β-alanine dose seems to be the most influential factor moderating muscle carnosine response to supplementation [16, 33], although the dose–response relationship tends to saturation [16, 46]. While the effects of β-alanine on muscle carnosine accretion are little influenced by factors such as sex and body mass [47], it is still controversial whether training status could affect muscle carnosine responses to β-alanine [38, 48]. Muscle carnosine content was shown to return to pre-supplementation levels 12–16 weeks after the cessation of β-alanine supplementation [33, 37]. It has been suggested that a lower maintenance dose (~ 1.2 g.d^{-1}) could be used after the initial loading to maintain muscle carnosine elevated for prolonged periods of time [47]. Exercise duration, and thus exercise intensity, seems to be the most influential factor moderating the efficacy of β-alanine on performance. While β-alanine is ergogenic in exercise tasks of 0.5–10 min of duration, increasing the performance by 0.2%–3% [43], no beneficial effects are observed in very short duration exercises (< 0.5 min) [12]. In addition, β-alanine supplementation seems to be more effective in non-trained than in trained individuals [12], although a study specifically designed to address this question showed similar ergogenic responses between well-trained athletes and untrained individuals [49].

β-Alanine side effects

Paresthesia

Oral β-alanine doses larger than 800 mg commonly result in unpleasant sensory symptoms [10]. However, there are no adverse events reported to β-alanine usage, only side effects such as paresthesia [13, 50]. Paresthesia is the only side effect of β-alanine supplementation that has been shown to occur in humans [13]. Described as an uncomfortable itchy or pins-and-needles sensation on the skin, paresthesia occurs within ~ 10–20 min following β-alanine ingestion [10] and is closely related with plasma β-alanine concentrations [51]. The mechanisms underpinning paresthesia seem to involve the Mas-related gene family of G-protein-coupled receptors, which are triggered by interactions with ligands including β-alanine [52]. Symptoms typically arise from a high dose of β-alanine with its intensity being closely associated with β-alanine concentrations in blood [47]. This means that paresthesia can be avoided if β-alanine is taken in the form of slow-release tablets [51] or if smaller and more frequent doses are ingested [51].

In a review about β-alanine safety specifically related to military personnel [50], β-alanine intervention studies and review articles were collected from 13 databases, and safety information was collected from adverse event reporting portals. Safety information from clinical studies indicated paresthesia as a short-term adverse reaction. However, the strength of evidence for a causal relationship between β-alanine and paresthesia was moderate. The overall strength of the evidence concerning the use of β-alanine and its relation to paresthesia came from the dose–response correlation, temporal correlation of the effect, high frequency of occurrence in the exposed population, correlation to plasma levels (and lack of correlation following use of a sustained-release formulation).

Because paresthesia does not necessarily impose any health risk per se, it can be considered a side effect instead of an adverse effect. However, given the relatively common reports of paresthesia in short-term studies, and how little is known about the physiological consequences thereof, it is important to monitor this over a prolonged follow-up period to determine whether increased incidence or symptom aggravation occurs as a function of time. Nevertheless, the best evidence to date suggests that β-alanine has a positive

safety profile for human consumption, at least when taken within the recommended doses and for up to 6 months [13]. More studies are needed to confirm whether longer periods of supplementation are equally safe.

Intramuscular taurine depletion

Taurine is a β-sulfonic amino acid present in human tissues and participates in a variety of important biological processes, including osmoregulation, cation homeostasis, enzyme activity, receptor regulation, cell development and signaling, and cytoprotective activity [53, 54]. Intramuscular taurine depletion is often a concern for individuals under chronic β-alanine supplementation. The trans-sarcolemmal transport of β-alanine into muscle is predominantly mediated by TauT [55], a sodium- and chloride-dependent transporter [56] that also transports taurine into muscle. β-Alanine and taurine, therefore, compete for the same transporter, and so the increases in β-alanine concentration induced by supplementation could, at least hypothetically, affect the transport of taurine, reducing taurine concentration within the skeletal muscle. However, a recent meta-analysis showed no effect of β-alanine supplementation on skeletal muscle taurine [13], which shows that β-alanine supplementation is safe when ingested within the doses used in the available studies.

Pruritus or Itching

The reports from WHO-UMC [57] present a signal for pruritus or itching following β-alanine consumption. These reactions were not observed in clinical studies, and the molecular mechanism for the reactions is not known. Pruritus involves a histamine-mediated allergic reaction and is not certain whether increased carnosine (beta-alanyl-L-histidine) levels following β-alanine consumption influences precursor availability for histamine production and release (as shown in cell culture in animal studies) [58, 59]. However, such possibility seems unlikely because a controlled-release formulation of β-alanine that increased muscle carnosine levels was not associated with symptoms of pruritus [33].

Evidence-based systematic review and meta-analysis evaluating the safety of β-alanine

Due to the high popularity of β-alanine [60], it is vital to determine whether supplementation within recommended doses could lead to any adverse outcomes. The International Society of Sports Nutrition stands by the position that β-alanine supplementation currently appears to be safe in healthy populations at recommended doses and the only reported side effect is paresthesia, which can be avoided by reducing single doses or using sustained-release formulas [61]. Nevertheless, the long-term effects of β-alanine interventions are unknown, and researchers should evaluate cumulative long-term effects [50]. The Australian Institute of Sport provides the expertise and resources developed during the implementation of the AIS Sports Supplement Program (2000–2013) to allow National Sporting Organizations and agencies to develop their own Sports Supplement Guidelines and Programs. The AIS Sports Supplement Framework has established β-alanine as a performance supplement with strong scientific evidence for use in specific situations in sport using evidence-based protocols. Therefore, β-alanine was classified as evidence level grade A in the ABCD system ranks sports foods and supplement ingredients, according to its scientific evidence and other practical considerations that determine whether a product is safe, legal and effective in improving sports performance. However, it is best used with an individualized and event-specific protocol, with the expert guidance of an accredited sports dietitian [62].

Since β-alanine is predominately metabolized in the liver and kidneys [32], it is important to determine whether chronic supplementation could overburden these organs. However, a recent systematic review of the safety profile of β-alanine supplementation reported no detrimental effects on a series of health markers, including kidney and hepatic function [13]. β-Alanine supplementation only caused a small increase in circulating alanine aminotransferase concentration (effect size, ES: 0.274, CrI: 0.04, 0.527), although mean data remained within clinical reference ranges [13]. Another study showed no adverse effect on 12-lead electrocardiogram, clinical chemistry or hematological safety data after 4 weeks of β-alanine supplementation at 3.2 g day^{-1} [10]. This was confirmed by Stellingwerff et al. [33] after 4 weeks with 3.2 $g.day^{-1}$, followed by 4 weeks of 1.6 g day^{-1}, and a total 8 weeks of 1.6 $g.day^{-1}$. Twelve weeks of supplementation at 3.2 g day^{-1} in older adults (60–80 years old) also showed no adverse effects on clinical health markers [63]. However, these studies used relatively low doses ($\leq$3.2 $g.day^{-1}$) and short supplementation periods ($\leq$12 weeks).

No studies have evaluated the effects of β-alanine supplementation for more than 6 months; hence, the possibility of longer-term adverse effects cannot be ruled out. Considering that a chronic and high-dose approach to supplementation is likely to be taken by certain athletic populations [60], it is important to determine the safety profile over longer periods of time.

Future directions

The available literature provides limited evidence concerning the side effects of β-alanine with the majority of the studies not being designed to address safety. In general, the

clinical studies did not use safety parameters as primary end points. A number of β-alanine studies are not well designed for safety outcomes, missing critical information. Future research should address confounders and insufficiency of evidence identified in the current review as the research gaps (e.g., risk of bias, sample size, and missing information). Furthermore, large cohort studies with multivariate analyses are needed to fully understand the benefits and safety of β-alanine among representative populations to understand the effects of variables such as sex, age, and ethnicity [50]. Future studies should evaluate the side effects of β-alanine supplementation for more than 6 months to adequately assess whether there are any long-term side effects.

Conclusion

The available literature indicates no adverse events reported for β-alanine usage, and side effects such as paresthesia may be observed if a single high dose of β-alanine is taken. Paresthesia can be attenuated either by splitting doses (< 800 mg) or by using sustained-release formulas. However, the adverse effects of long-term supplementation (> 6 months) at recommended doses are still unknown. β-Alanine is currently considered to be safe in healthy populations at recommended doses.

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Declarations

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Ethical approval Not applicable.

Informed consent Not applicable.

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